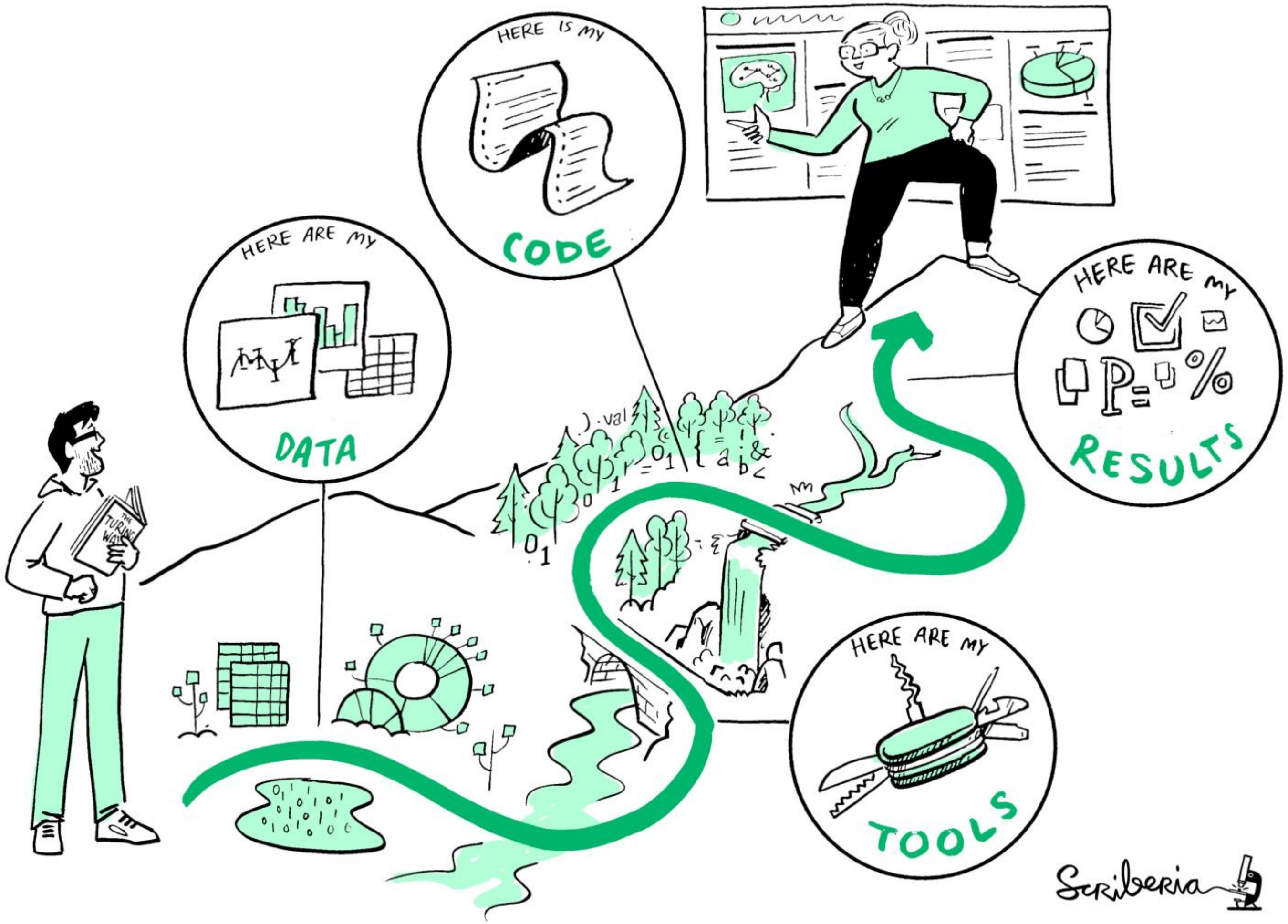
Reproducible Manuscripts with Quarto

Shreya Dimri

(EcoOpen Symposium for RQI)

What is meant by reproducible here?



The Turing Way project illustration by Scriberia

But your research is often more than a PDF, it is a “knowledge stack”

The Turing Way project illustration by Scriberia

How do we write manuscripts at the moment?

- What do you use?
- And why?

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- What do you use?
- And why?
- The usual methods can have serious drawbacks i.e. barriers to reproducibility
- Re-submission cycle with painstaking reformatting

What do we want?

Dynamic Document Generation, where we can:

- combine narrative with code

What do we want?

Dynamic Document Generation, where we can:

- combine narrative with code
- automatically generate figures and tables
- automatically render results in text
- format the content into a scientific paper (including citations!)

What do we want?

Dynamic Document Generation, where we can:

- combine narrative with code
- automatically generate figures and tables
- automatically render results in text
- format the content into a scientific paper (including citations!)
- Repeat it whenever needed!

Benefits of a dynamic document

Eliminate human error in copying and pasting results

“We found that half of all published psychology papers that use null-hypothesis significance testing (NHST) contained at least one p-value that was inconsistent with its test statistic and degrees of freedom. One in eight papers contained a grossly inconsistent p-value that may have affected the statistical conclusion”

Nuijten et al., 2016

Benefits of a dynamic document

Easy revisions and specification of desired figures and tables

Promote computational reproducibility

- Easy verification and replication of research findings
- Ultimately reduce mistakes and maybe even save time (once you climb the mount learning)

What is Quarto?

- **Open-source** publishing system (successor to R Markdown)
- Allows for literate programming in Markdown and with integrated tools making reproducibility a default

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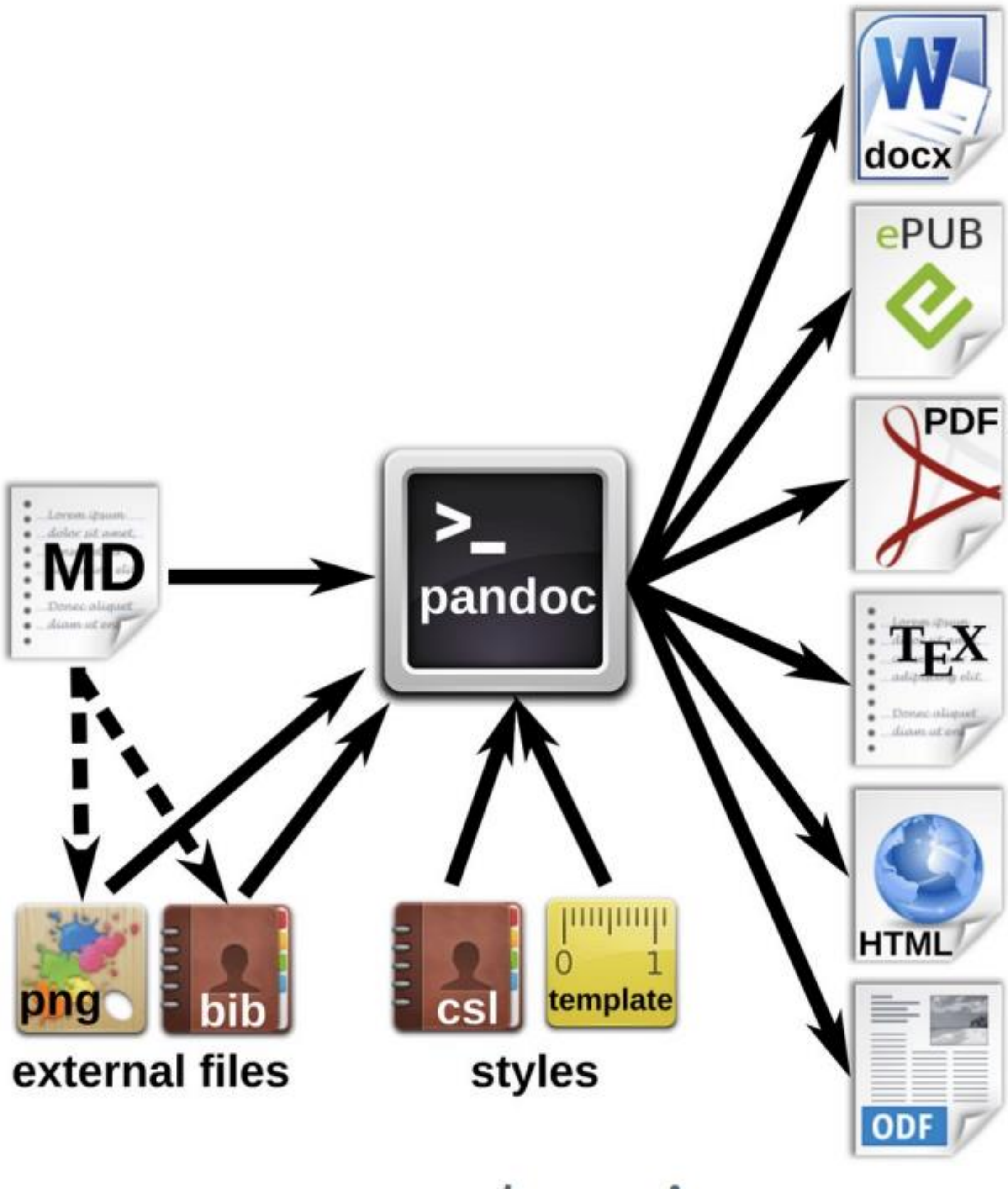
- Versatility (Choice of editors and output formats)
- Supports R, Python, Julia, ObservableJS.. and potentially more in the future
- Exports to HTML, PDF, Word, books, slides, websites.. over 40 possible formats

What is Quarto?

- **Open-source** publishing system (successor to R Markdown)
- Allows for literate programming in Markdown and with integrated tools making reproducibility a default
- Versatility (Choice of editors and output formats)
- Supports R, Python, Julia, ObservableJS.. and potentially more in the future
- Exports to HTML, PDF, Word, books, slides, websites.. over 40 possible formats
- Reference management and Cross referencing is possible



Authoring software options:

- RStudio
- JupyterLab
- VS Code

Components:

- YAML header (metadata)
- Markdown text
- Code chunks + inline code
- Bibliography

YAML header (metadata)

_quarto.yml

```yaml
project:
  type: manuscript
  output-dir: docs

manuscript:
 article: manuscript_GNM.qmd

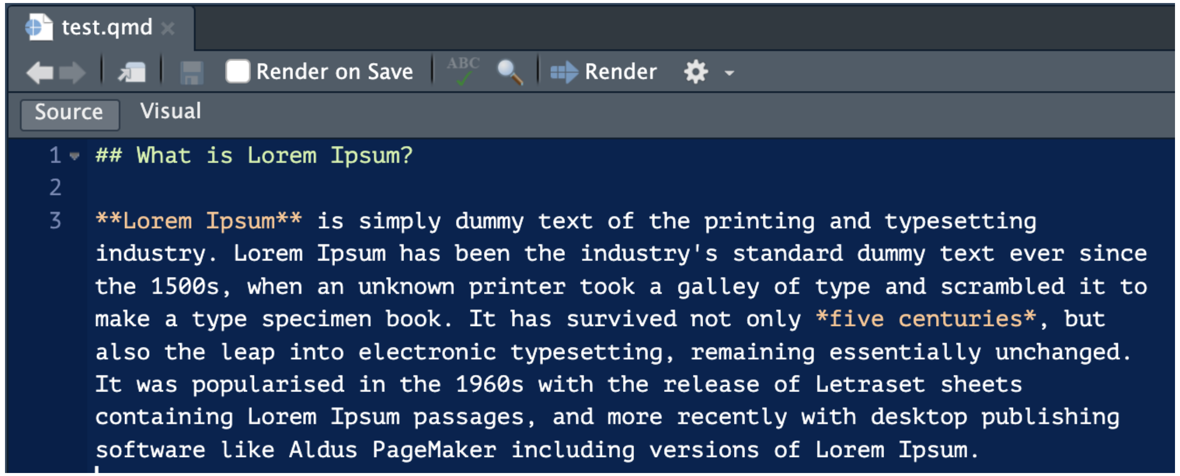

format:
 html:
  comments:
   hypothesis: true
  bibliography: references.bib
  csl: "proceedings-of-the-royal-society-b.csl"
 # docx:
 #  fontsize: 12
 # pdf: default


editor: visual
```

Markdown Text

What is Lorem Ipsum?

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.



Markdown Text

Markdown Syntax	Output
`*italics* and **bold**`	*italics* and **bold**
`superscript^2^ / subscript~2~`	superscript^2 / subscript$_2$
`~~strikethrough~~`	~~strikethrough~~
`` `verbatim code` ``	`verbatim code`

Markdown Text

Markdown Syntax	Output
`# Header 1`	**Header 1**
`## Header 2`	**Header 2**
`### Header 3`	**Header 3**
`#### Header 4`	**Header 4**

```
* unordered list
  + sub-item 1
    - sub-sub-item 1
  + sub-item 2
```

- unordered list
  - sub-item 1
    - sub-sub-item 1
  - sub-item 2

```
1. ordered list
2. item 2
   i. sub-item 1
      A.  sub-sub-item 1
```

1. ordered list
2. item 2
   i. sub-item 1
      a. sub-sub-item 1

```
*   item 2

    Continued (indent 4 spaces)
```

- item 2

  Continued (indent 4 spaces)

```
term
: definition
```

term
  definition

Markdown Text

```
| Right | Left | Default | Center |
|------:|:-----|---------|:------:|
|   12  |  12  |    12   |    12  |
|  123  |  123 |   123   |   123  |
|    1  |    1 |     1   |     1  |
```

Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1

Markdown Text

```
You can embed [named hyperlinks](https://quarto.org/)

Simply a direct url like <https://quarto.org/>

Links to [other places](#your_link) in the document.

Or embed an image:
![cat](https://upload.wikimedia.org/wikipedia/commons/4/4d/Cat_November_2010-1a.jpg)
```

You can embed named hyperlinks

Simply a direct url like https://quarto.org/

Links to other places in the document.

Or embed an image:

Markdown Text

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![cat](https://upload.wikimedia.org/wikipedia/commons/4/4d/Cat_November_20
10-1a.jpg)
```

You can embed named hyperlinks

Simply a direct url like https://quarto.org/

Links to other places in the document.

Or embed an image:

https://www.markdownguide.org

References/Bibliography

A typical BibTex key might look like as follows:

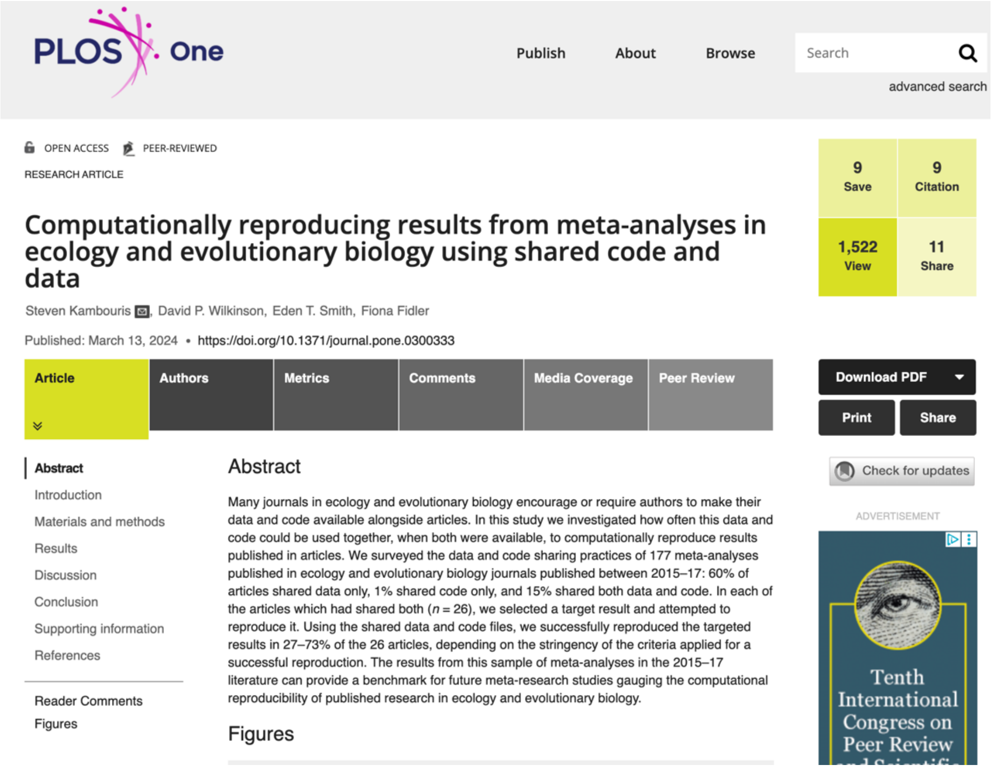
```
@article{10.1371/journal.pone.0300333,
    doi = {10.1371/journal.pone.0300333},
    author = {Kambouris, Steven AND Wilkinson, David P. AND Smith, Eden T. AND
Fidler, Fiona},
    journal = {PLOS ONE},
    publisher = {Public Library of Science},
    title = {Computationally reproducing results from meta-analyses in ecology and
evolutionary biology using shared code and data},
    year = {2024},
    month = {03},
    volume = {19},
    url = {https://doi.org/10.1371/journal.pone.0300333},
    pages = {1-22},
    number = {3},

}
```

Referencing workflow (example)

My typical workflow is as follows:

- Find a reference that one needs in your manuscript.

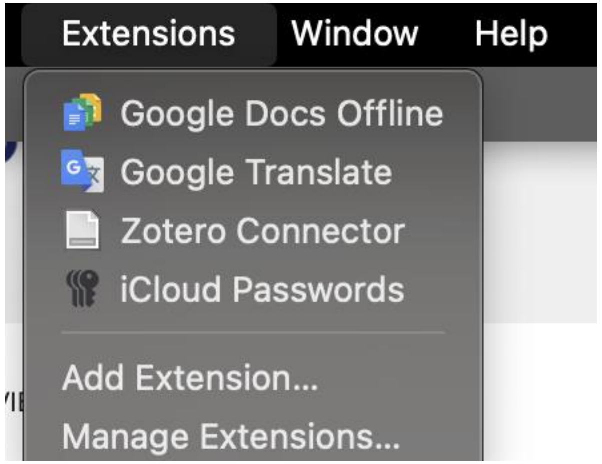


Referencing workflow (example)

My typical workflow is as follows:

- Find a reference that one needs in your manuscript.
- Save it to your Zotero library via the browser extension

Zotero automatically updates your .bib file through syncing with Zotero desktop and updating the .bib file.



Referencing workflow (example)

- Cite the references the @ + reference identifier:
  - **In-text citations**:
  - **Bracketed citations**:

```
Now you can cite the article in the text you are writing @kambouris2024

Now you can cite the article in the text you are writing [@kambouris2024]
```

Now you can cite the article in the text you are writing Kambouris et al. (2024)

Now you can cite the article in the text you are writing (Kambouris et al. 2024)

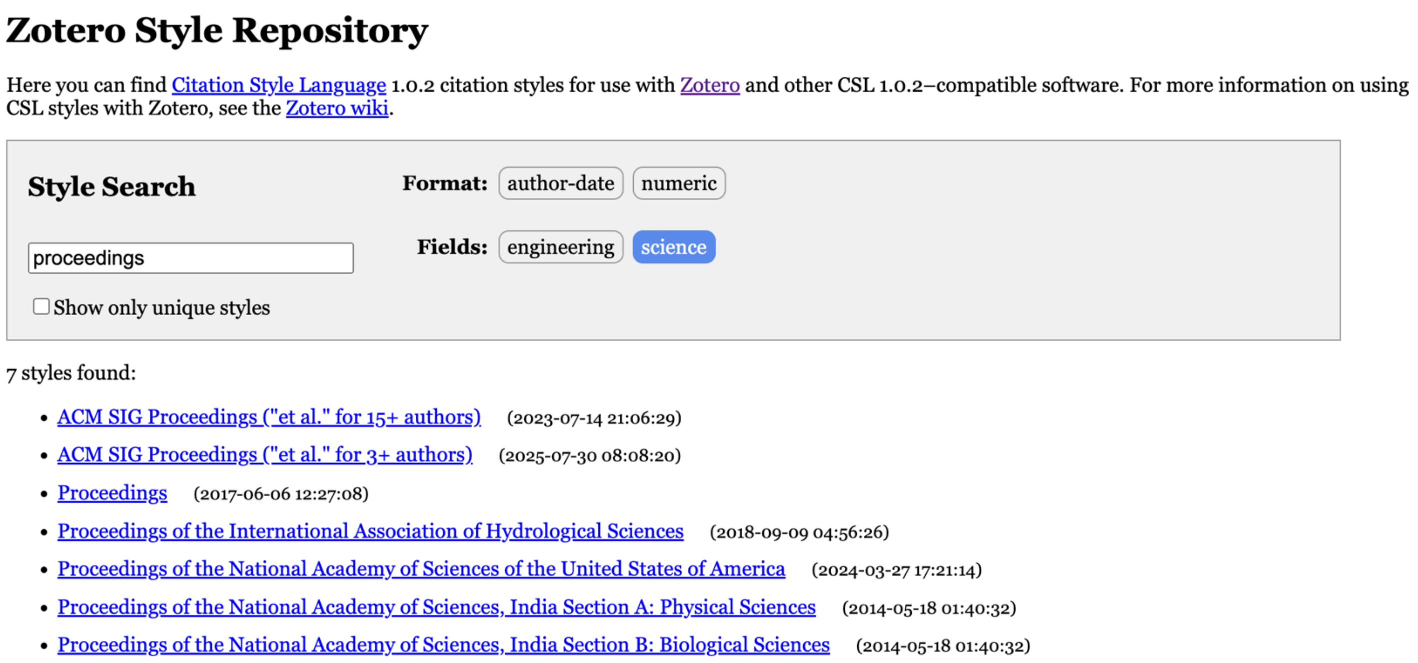
References

Kambouris, Steven, David P. Wilkinson, Eden T. Smith, and Fiona Fidler. 2024. "Computationally Reproducing Results from Meta-Analyses in Ecology and Evolutionary Biology Using Shared Code and Data." *PLOS ONE* 19 (3): e0300333. https://doi.org/10.1371/journal.pone.0300333.

References/Bibliography

Switching referencing style according to journals:

- Search your style of preference: https://www.zotero.org/styles



Zotero Style Repository

Here you can find Citation Style Language 1.0.2 citation styles for use with Zotero and other CSL 1.0.2–compatible software. For more information on using CSL styles with Zotero, see the Zotero wiki.

Style Search

Format: author-date numeric

Fields: engineering science

proceedings

Show only unique styles

7 styles found:

- ACM SIG Proceedings ("et al." for 15+ authors) (2023-07-14 21:06:29)
- ACM SIG Proceedings ("et al." for 3+ authors) (2025-07-30 08:08:20)
- Proceedings (2017-06-06 12:27:08)
- Proceedings of the International Association of Hydrological Sciences (2018-09-09 04:56:26)
- Proceedings of the National Academy of Sciences of the United States of America (2024-03-27 17:21:14)
- Proceedings of the National Academy of Sciences, India Section A: Physical Sciences (2014-05-18 01:40:32)
- Proceedings of the National Academy of Sciences, India Section B: Biological Sciences (2014-05-18 01:40:32)

References/Bibliography

Switching referencing style according to journals:

- Search your style of preference: https://www.zotero.org/styles
- Download the csl file into your manuscript folder
- Reference the csl file in your yaml:

```yaml
project:
  type: manuscript
  output-dir: docs

manuscript:
 article: manuscript_GNM.qmd


format:
 html:
  comments:
   hypothesis: true
  bibliography: references.bib
  csl: "proceedings-of-the-royal-society-b.csl"
 # docx:
 #  fontsize: 12
 # pdf: default


editor: visual
```

Now things become interesting with code

Inline code: Insert results or (quick) calculations into running text, without additional text formatting.

- report and describe results reproducibly
- minimizes errors: no manual copy-pasting results
- automatic updating when data or code changes

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```
{r echo=F}
x <- 5  # radius of a circle

For a circle with the radius `r x`, its area is `r pi * x^2`.
```

For a circle with the radius 5, its area is 78.5398163.

Source your code

Sourcing external files: For example: R and Python scripts, .qmd or .ipynb files.

Why separate scripts from the manuscript?

- minimizes errors: no manual copy-pasting code
- easier code debugging
- making the Quarto document easier to digest
- modular coding = good coding practice

Source your code

```
{r echo=F}
x <- 5  # radius of a circle
```

For a circle with the radius `r x`, its area is `r pi * x^2`.

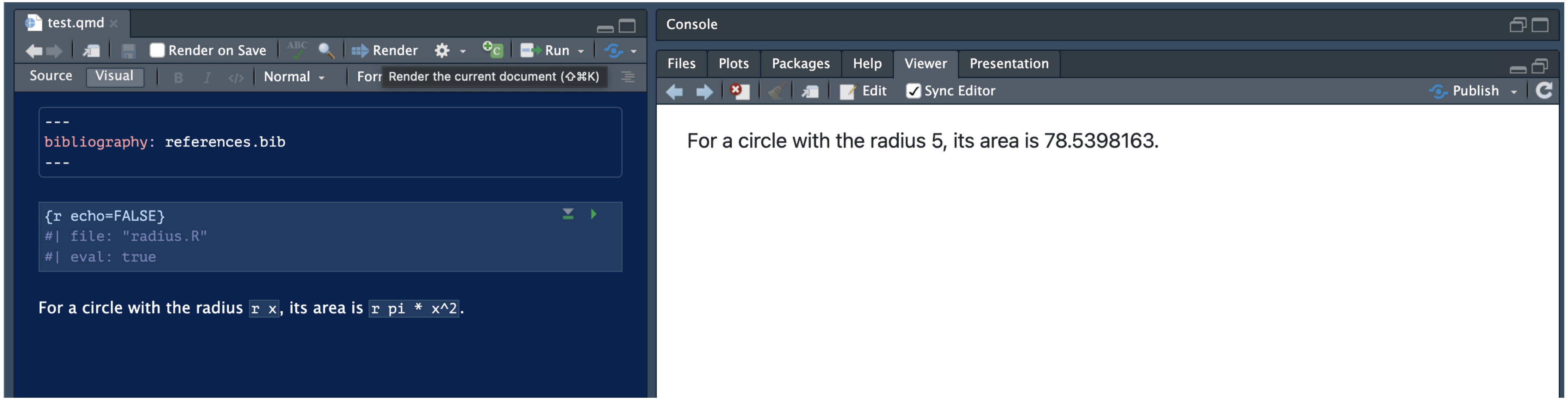
For a circle with the radius 5, its area is 78.5398163.

```
{r echo=FALSE}
#| file: "radius.R"
#| eval: true
```

For a circle with the radius `r x`, its area is `r pi * x^2`.

For a circle with the radius 5, its area is 78.5398163.

Render your document

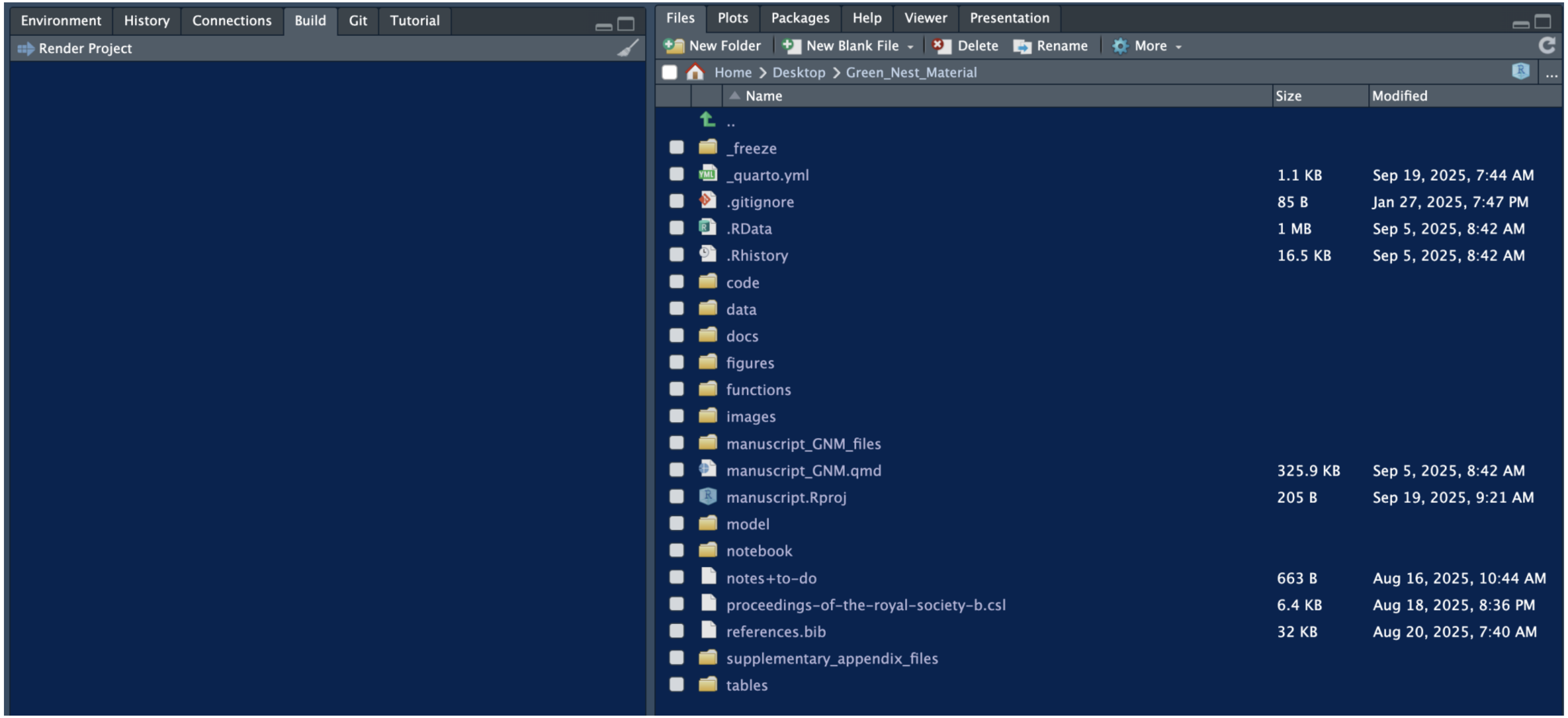


Render your document

```r
library(quarto)
quarto_render("my-reproducible-manuscript.qmd") # defaults to html
quarto_render("my-reproducible-manuscript.qmd", output_format = "pdf")
```

```bash
quarto render
quarto render my-reproducible-manuscript.qmd # defaults to html
quarto render my-reproducible-manuscript.qmd --to pdf
```

Render your document



Render your document

Table of contents

Notebooks

For the few cases for which we could not extract means and variances but inferential statistics such as t and F values, we estimated SMD values (k = 4, n = 2) using the equations provided by Cooper, Hedges, and Valentine (2019). For dichotomous variables (e.g., number of successful nests, k = 4, n = 2), we calculated the log odds ratio (lnOR) using the 'escalc' function (measure=OR) from 'metafor' and subsequently converted it to SMD using equations from Cooper, Hedges, and Valentine (2019). Since lnRR cannot be calculated from inferential statistics, dichotomous variables, or traits including zero or negative values (k = 7), the number of effect sizes included in the lnRR (k = 259, n = 28) subset is smaller than that of SMDH (k = 274, n = 28).

When calculating each effect size's associated sampling variance, we prioritised the number of nests in each group over the number of individuals as the sample size whenever possible to reduce the effect of genetic relatedness non-independence. In cases where the same control or treatment group was used across multiple comparisons, sample sizes were adjusted prior to calculating effect sizes by dividing them by the number of times each group was used in a comparison to account for shared group non-independence. For cases where the authors of an original article reported a statistically non-significant difference between two groups but without providing numerical values (k = 16, n = 2), we assigned zero as the effect size for that comparison and imputed its sampling variance using the missing-case approach described in Nakagawa, Noble, et al. (2022) for lnRR and using $\frac{n_1+n_2}{n_1 \times n_2}$ for SMD (Lajeunesse 2013).

Data analysis

We performed all analyses in R v4.5.1 (R Core Team 2025). All multilevel meta-analytical models and meta-regressions were fitted using 'metafor' v.4.8-0 (Viechtbauer 2010). Sampling variances were modelled as variance-covariance matrices assuming a 0.5 correlation (ρ) between sampling variances from the same study (Noble et al. 2017) using the vcalc function from 'metafor' (See Supplementary Appendix Figure S8 for sensitivity analyses using different values of ρ).

Render your document

Table of contents

Notebooks

Results

We obtained 274 effect sizes from a total of 28 studies conducted on 7 species across 17 geographical locations (Figure1). About 81% of the effect sizes came from three bird species (Cyanistes caeruleus, k = 77; Sturnus unicolor, k = 60; Sturnus vulgaris, k = 86). Our dataset included 2 unpublished studies (k = 18), with the remaining studies being published between 1988 and 2024. The mean ±standard deviation number of effect sizes per study was 9.8 ± 6.4 (median: 8.5 range: 2–24) and the mean effective sample size (i.e., the number of nests/individuals per effect size after accounting for shared-group non-independence) was 32.3 ±31.1 (median: 22.0, range: 3–210; 31 data points had a sample size of less than 5 nests in each group).

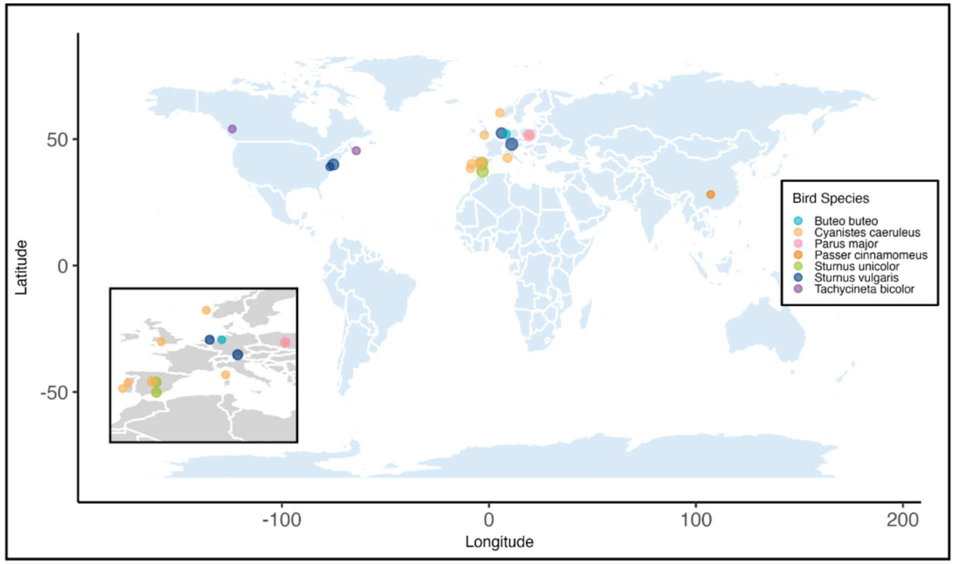


Figure1: **Geographical distribution of the 17 study populations included in a meta-analysis testing the adaptive value of green nest material in birds**.Point size reflects the number of effect sizes coming from the location (range: 3 - 34 effect sizes), and colour denotes species (see legend), with an inset zooming on a dense cluster of data points.

Render your document

Meta-analytic mean

Both lnRR and SMD(H) intercept-only multilevel meta-analyses showed an average increase in fitness estimates in nests with an experimental increase in GNM compared to control nests; however, the overall effect was only statistically significant for SMD(H).

First, lnRR showed an average statistically non-significant increase in fitness outcomes of 1.9% in nests with an experimental increase in GNM (Figure2 A; lnRR = 0.019, 95% CI: [-0.012 to 0.050], 95% PI: [-0.281 to 0.319], p-value = 0.223, k = 238, n = 26) but heterogeneity among effect sizes was high. Total raw heterogeneity (σ^2) was 0.023 (Q = 2277.7, p < 0.001), the measures of relative (I^2_{total} = 98.6%) and standardised heterogeneity ($CVH2_{total}$ = 63.59 and $M2_{total}$ = 0.98) suggested high heterogeneity across effect sizes.

Second, SMD(H) showed an average statistically significant increase in fitness outcomes in nests with an experimental increase of GNM (Figure2 B; SMD(H) = 0.179, 95% CI: [0.048 to 0.310], 95% PI: [-0.829 to 1.188], p-value = 0.008, k = 253, n = 26), but heterogeneity among effect sizes was moderate to high. Total raw heterogeneity (σ^2) was 0.258 (Q = 1114.7, p < 0.001), relative heterogeneity was smaller compared to that of lnRR (I^2_{total} = 66.1%) and the standardised heterogeneity metrics ($CVH2_{total}$ = 8.02 and $M2_{total}$ = 0.89) lied between the 50th and 75th percentile of empirically derived estimates from meta-analyses in ecology and evolution (Yang, Noble, et al. 2024). For lnRR, all three heterogeneity metrics exceeded the 75th

Limitations of Quarto

- Requires R/Python + (for PDF) LaTeX setup
- Learning curve (Markdown/YAML)
- Less familiar to collaborators/editors
- Journal templates sometimes need manual cleanup
- Smaller community (compared to LaTeX or Rmd)

short-term learning curve for long-term gain

Resources

https://utrechtuniversity.github.io/workshop-reproducible-manuscripts

https://lmu-osc.github.io/introduction-to-Quarto/

https://mine.quarto.pub/quarto-manuscripts-rmed/#/whats-next

https://quarto.org/docs/manuscripts/


No thanks!
We are too busy